Xiaoliang Wang | wxl989856@gmail.com | Medical AI / Multimodal Reasoning / RSI

Education

Zhejiang University, Electronic Science and Technology, *M.S.* (GPA: 91.59/100) 2024.09 - 2027.06
Zhejiang University, Electronic Science and Technology, *B.S.* (GPA: 3.88/4.0) 2020.09 - 2024.06

Publications and Manuscripts

- **Xiaoliang Wang**, et al. **MedEvo: Medical Reasoning Evolution through Knowledge Navigation**. *Submitted to ICLR 2027*.
- **Xiaoliang Wang**, et al. **ECG-RSI: Recursive Self-Improvement through Targeted ECG Waveform Evolution**. *Submitted to ICLR 2027*.
- **Xiaoliang Wang**, Xiang J, Tu F, et al. Detection and Localization of Early-Stage Abnormal Sounds in Lithium-Ion Battery Thermal Runaway. *Journal of Energy Storage*, 2025. (JCR Q1, IF = 10.7)
- **Xiaoliang Wang**, Ao M, Chen M, et al. LiB-TRAD: A Lithium Battery Thermal Runaway Acoustic Dataset for Anomaly Detection. (EI)

Research Experience

MedEvo: Medical Reasoning Evolution through Knowledge Navigation 2026.06 - 2026.10

Project page: https://pork-pay.github.io/MedEvo/

- Proposed **MedEvo**, a medical reasoning self-improvement framework that converts model failures into effective supervision through **knowledge-guided trajectory repair and feedback-driven data evolution**.
- Developed a **gold-blind local trajectory repair** pipeline that identifies the first consequential reasoning error, preserves the valid reasoning prefix, and guides the same student model to regenerate the erroneous suffix using minimal answer-free hints.
- Designed a four-dimensional medical taxonomy over **modality, anatomy, disease, and difficulty**, together with a **20-subtype medical error codebook**, to diagnose capability gaps on a held-out feedback set and dynamically prioritize questions and failure types across self-training rounds.
- Using SFT alone, improved **Qwen3.5-9B** from **65.57% to 69.12%** averaged over seven medical multimodal benchmarks and from **47.40% to 54.50%** on MedXpertQA-MM. The framework further improved Qwen3.5-35B from **67.91% to 73.04%** and Qwen3.5-397B from **71.83% to 74.95%**.

ECG-RSI: Recursive Self-Improvement through Targeted ECG Waveform Evolution 2026.07 - 2026.10

Project page: https://pork-pay.github.io/ECG-RSI/

- Proposed **ECG-RSI**, a recursive self-improvement framework that connects learner feedback with **targeted ECG waveform synthesis**, enabling training data and supervision strategies to evolve with model-specific capability gaps.
- Developed measurement-verified waveform interventions over ECG **rhythm, intervals, morphology, amplitude, and lead-specific patterns**. Edited waveforms are automatically remeasured to verify target attainment and preserve non-target physiological properties before supervision is generated.
- Designed a **dual-feedback evolution loop**: construction failures refine waveform operators and generation strategies, while learner errors, difficulty profiles, and confusion patterns guide subsequent data recipes and same-parent intervention search.
- Across three RSI rounds, improved **Qwen3.5-35B-A3B** from **30.15 to 46.61** in aggregate score across seven ECG benchmarks. MedXpert ECG accuracy increased from **51.57% to 57.24%**, while LUDB accuracy increased from **53.45% to 67.24%**.

Acoustic Detection and Localization for Battery Thermal Runaway 2024.09 - 2025.11

- Constructed **LiB-TRAD**, a four-channel acoustic dataset covering lithium-ion battery thermal runaway, including **thermal-runaway recordings** and **normal-charging recordings**, with abnormal acoustic events manually annotated.
- Developed a **BiGRU-based temporal fusion model** for detecting weak and transient abnormal acoustic signals before safety-valve opening, achieving over **90% recognition accuracy** and enabling warnings up to **101 s before valve opening** and **791 s before full thermal runaway**.

- Designed a four-microphone localization system based on **TDOA** and **Adaptive Frequency-Weighted GCC**, achieving **90.21% localization accuracy**, a mean angular error of **4.39 degrees**, and approximately **0.05 m** spatial localization accuracy.

Research Internship

Ant Group - Afu Multimodal **Research Intern** 2026.06 - Present

- Conduct research on **medical multimodal reasoning and recursive self-improvement**, spanning supervised fine-tuning, reinforcement learning, failure-driven supervision, and adaptive medical data evolution.
- Led the development of **MedEvo** and **ECG-RSI**, investigating how model failures and capability feedback can be converted into new reasoning supervision and targeted multimodal training data across iterative self-improvement rounds.
- Built large-scale multimodal post-training pipelines for Qwen3.5-series models, covering distributed **SFT**, model rollout, trajectory-level error diagnosis, synthetic-data verification, feedback-guided data selection, and multi-benchmark evaluation.
- Developed a **DAPO-based reinforcement learning** pipeline for multimodal medical reasoning, using verifiable task rewards to optimize reasoning policies beyond supervised fine-tuning and investigate the combination of **SFT-based data evolution and RL-based policy optimization**.

Simplex AI **Research Intern - Agentic Search** 2026.03 - 2026.05

- Developed an **agentic reinforcement learning** framework using verl and GRPO, enabling language models to interleave multi-step reasoning with autonomous search actions. The resulting model improved HotpotQA Exact Match from **0.201 to 0.278**.
- Designed a structured **Search Plan / TaskItem** formulation for decomposing complex queries into specialized retrieval subtasks, including semantic retrieval, filtering, shortcut retrieval, and entity expansion.
- Built a multi-route neural search framework with **DB, Wide, Shortcut, and Entity Expansion** retrieval paths and dynamic result aggregation, improving search quality from **55% to 83% NDCG@20**.

Research Skills

- **LLM/VLM Post-Training:** Supervised fine-tuning, self-training, trajectory repair, recursive self-improvement, synthetic data generation and selection, and agentic reinforcement learning.
- **Medical AI and Multimodal Reasoning:** Medical vision-language models, medical visual question answering, ECG interpretation, evidence-grounded reasoning, medical data construction, and multimodal model evaluation.
- **Deep Learning and Engineering:** PyTorch, Transformers, Qwen-series models, distributed MoE training, ms-swift, verl, large-scale model rollout, and distributed multimodal evaluation.
- **Signal Processing:** ECG signal processing, acoustic signal processing, TDOA/GCC localization, anomaly detection, and time-series modeling.

Honors and Awards

- Zhejiang Provincial Government Scholarship
- Zhejiang University Second-Class and Third-Class Scholarships
- Outstanding Student, Zhejiang University
- Outstanding Graduate, Zhejiang University
- National Third Prize, Alipay Mini Program Development Competition

Research Interests

- **Recursive Self-Improvement and Data Evolution:** Failure-driven self-improvement, self-generated supervision, adaptive data synthesis and selection, and iterative post-training of foundation models.
- **Medical Multimodal Reasoning:** Evidence-grounded medical reasoning, reasoning trajectory repair, medical vision-language models, and reliable multimodal supervision.
- **Medical Signal and Image Understanding:** Fine-grained visual grounding, ECG interpretation, physiological signal understanding, and multimodal representation learning.